

Barnabé Ledoux^{a,*}, Ryoka Kuwabara^b, Norikazu Ichihashi^c, Ryo Mizuuchi^b, David Lacoste^a

(a) Gulliver Laboratory, UMR CNRS 7083, PSL Research University, ESPCI, Paris F-75231, France

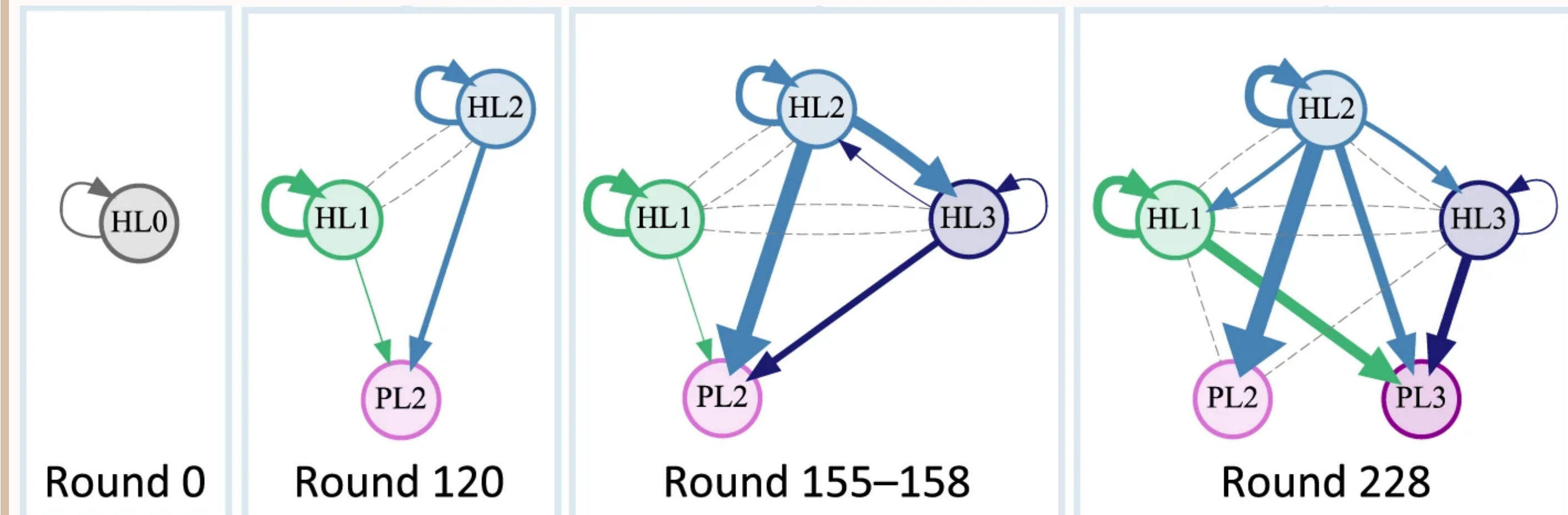
(b) Department of Electrical Engineering and Bioscience, Faculty of Science and Engineering, Waseda University, Shinjuku, Tokyo 162-8480, Japan

(c) Department of Life Science, Graduate School of Arts and Science, The University of Tokyo, Meguro, Tokyo 153-8902, Japan

*barnabe.ledoux@espci.fr

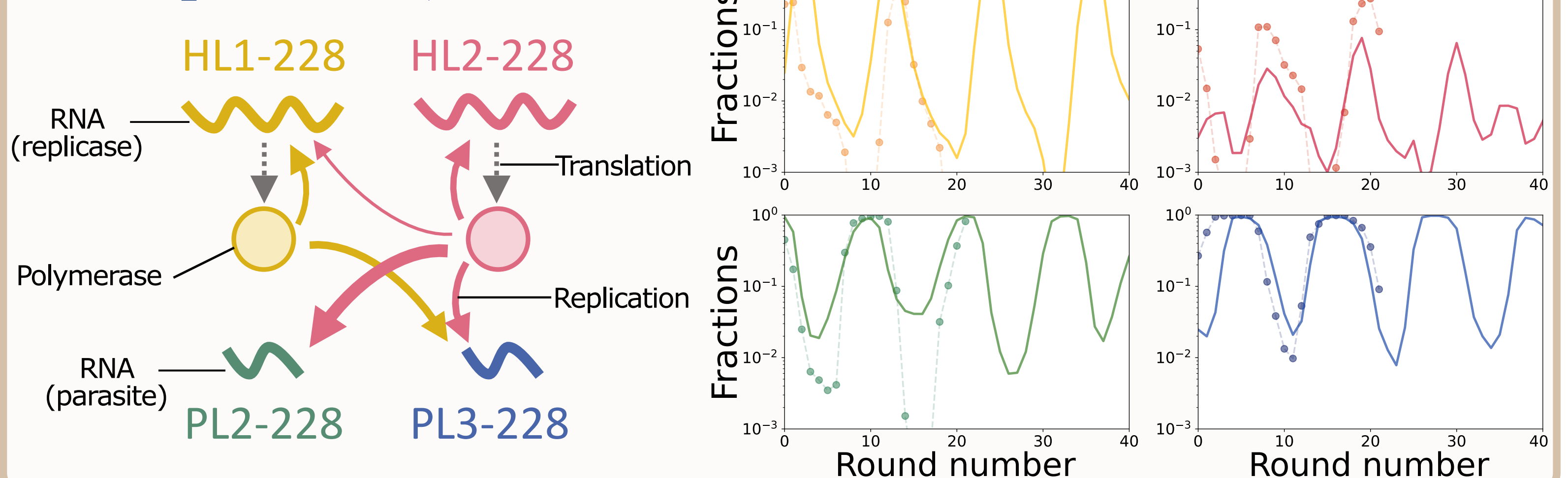
Evolution of complexity

Compartmentalization allows **longer and more complex sequences** to appear and survive against parasites [2].



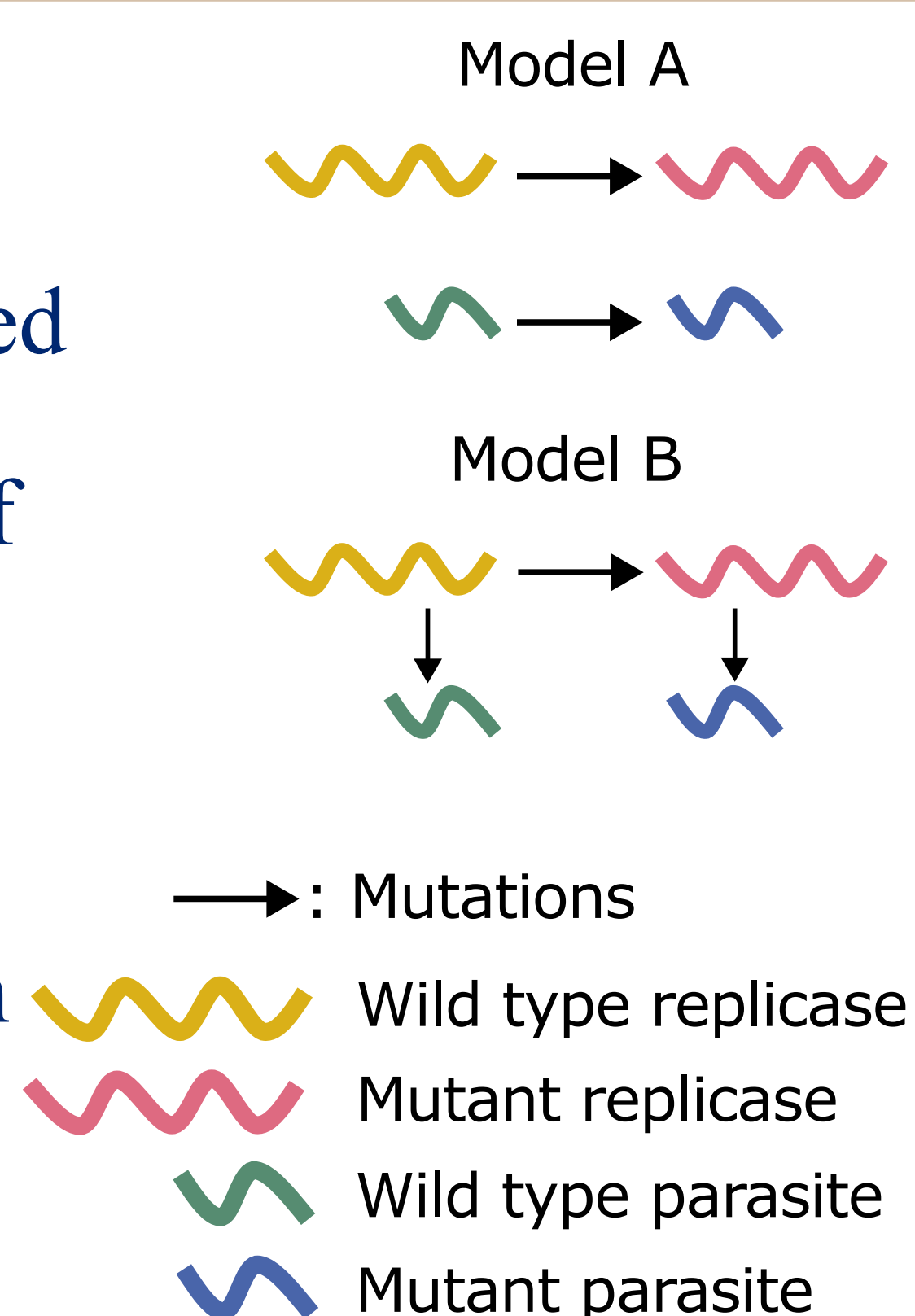
Experimental system

We observe **oscillations** between the different species (**theory and experiments**).



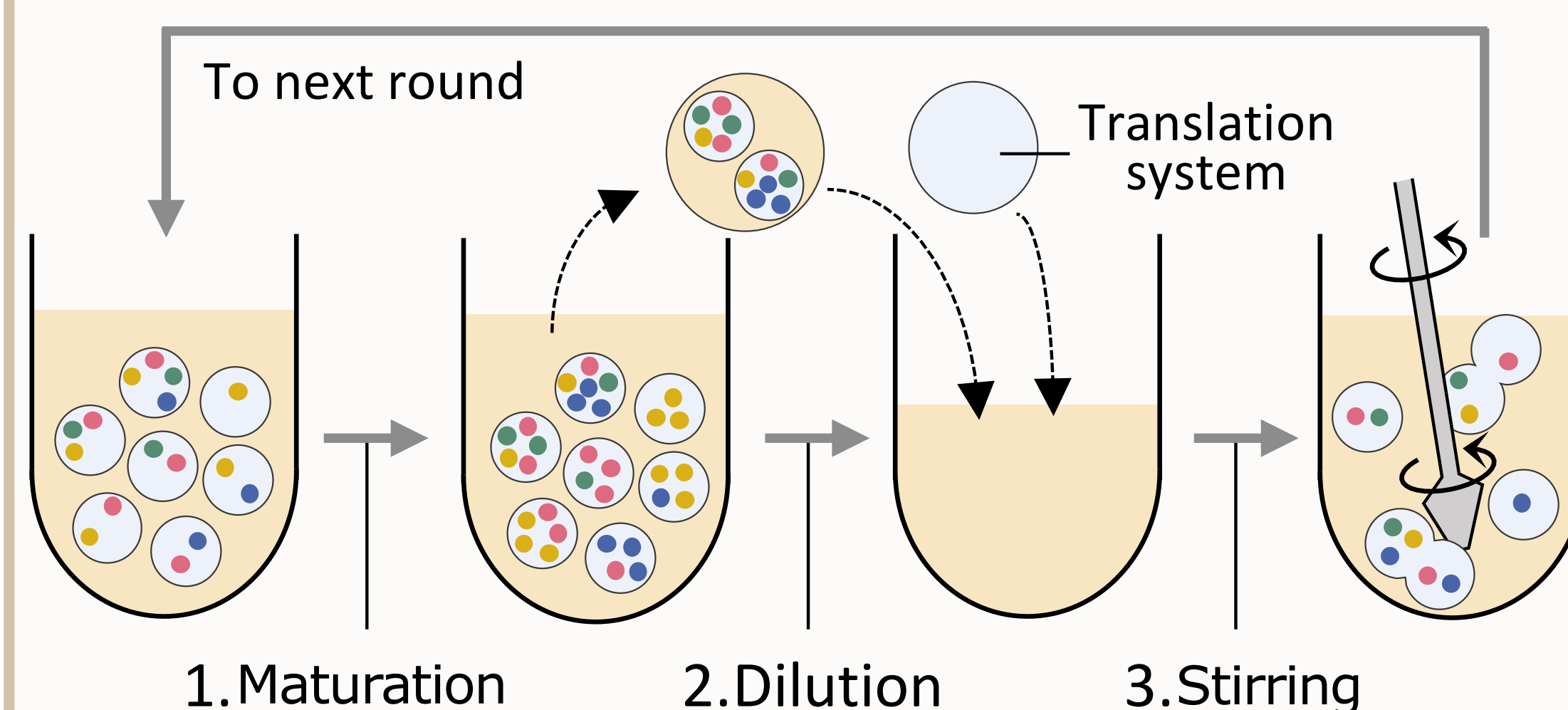
Models for maturation

We use a **deterministic approach** based on the initial composition of the compartments, which we satisfyingly compared with **Gillespie simulations**.



Transient compartmentalization with stirring

We add stirring to **transient compartmentalization dynamics** [2-4]



Maturation

$$\frac{dm_0^{(i)}}{dt} = -(\mu_0 + \nu_0)m_0^{(i)} + \alpha_{0,0}(m_0^{(i)})^2$$

Dilution

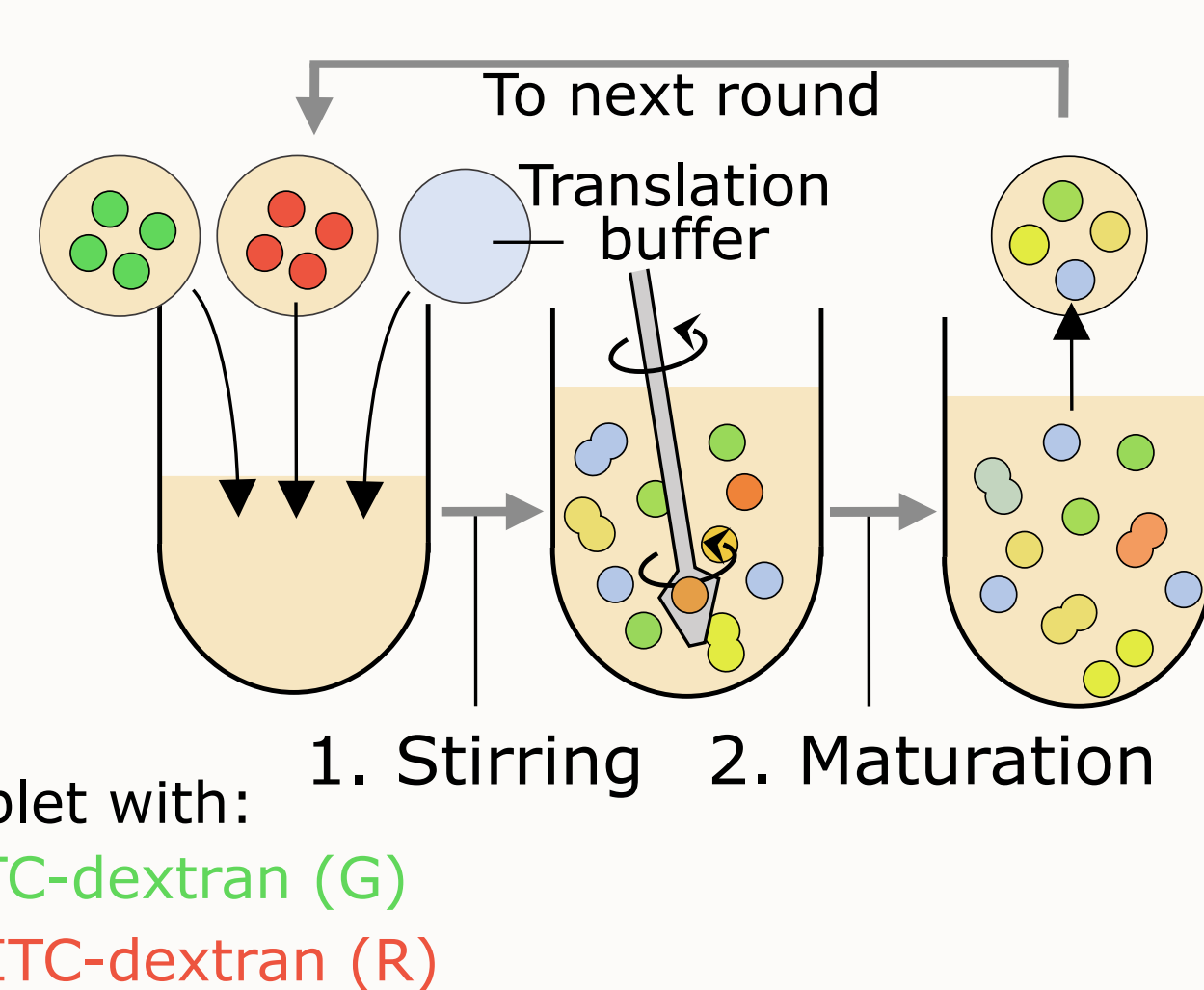
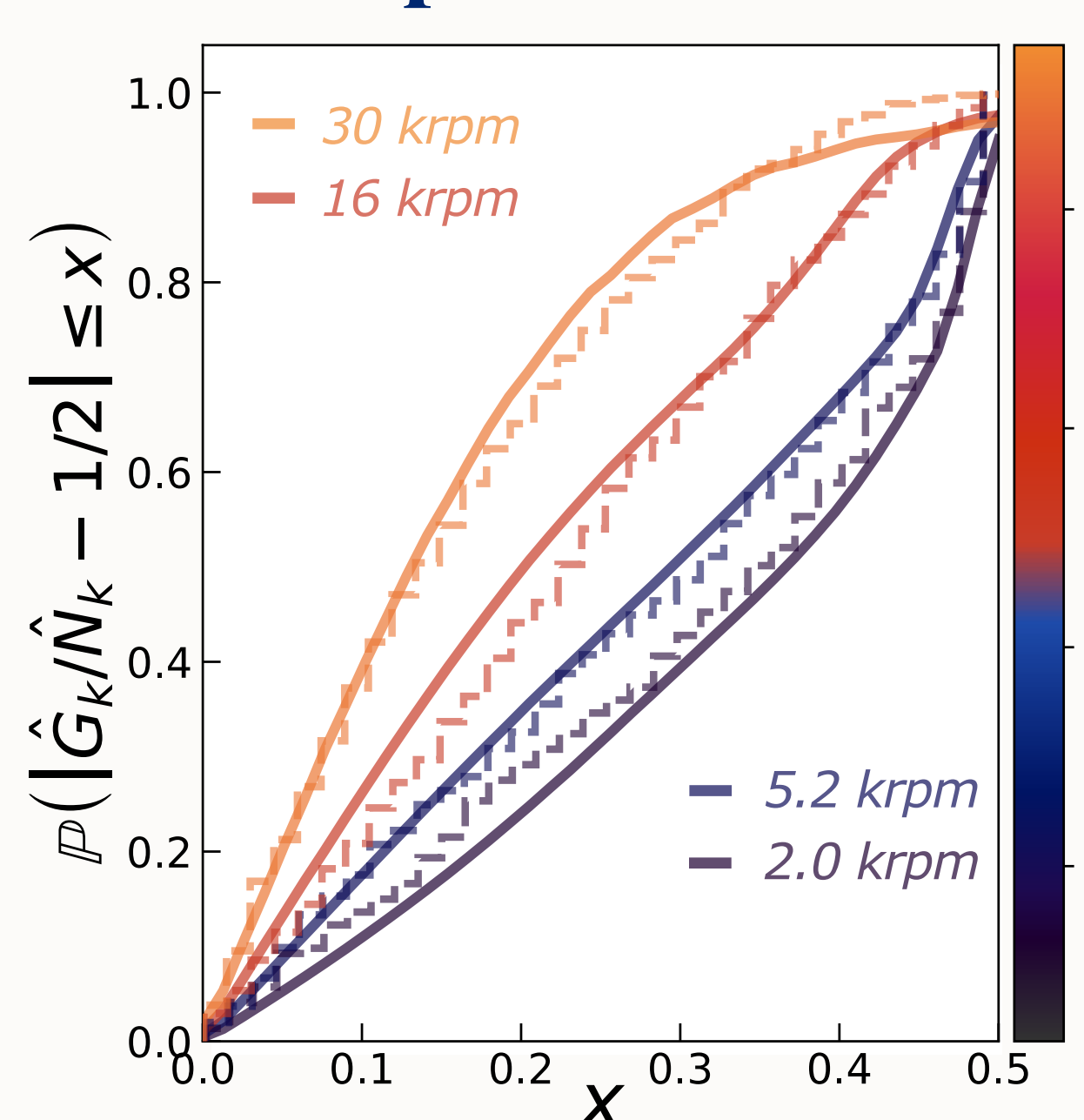
$$\langle \lambda^{(i)} \rangle = \frac{\langle m_0^{(i)} + m_1^{(i)} + y_0^{(i)} + y_1^{(i)} \rangle}{d}$$

Stirring (mean-field effect)

$$\hat{x}_0^{(i)} = \frac{(1-s)\bar{m}_0^{(i)} + s\langle \bar{m}_0 \rangle}{(1-s)\bar{n}^{(i)} + s\frac{\langle \bar{n} \rangle}{d}}$$

Measuring stirring

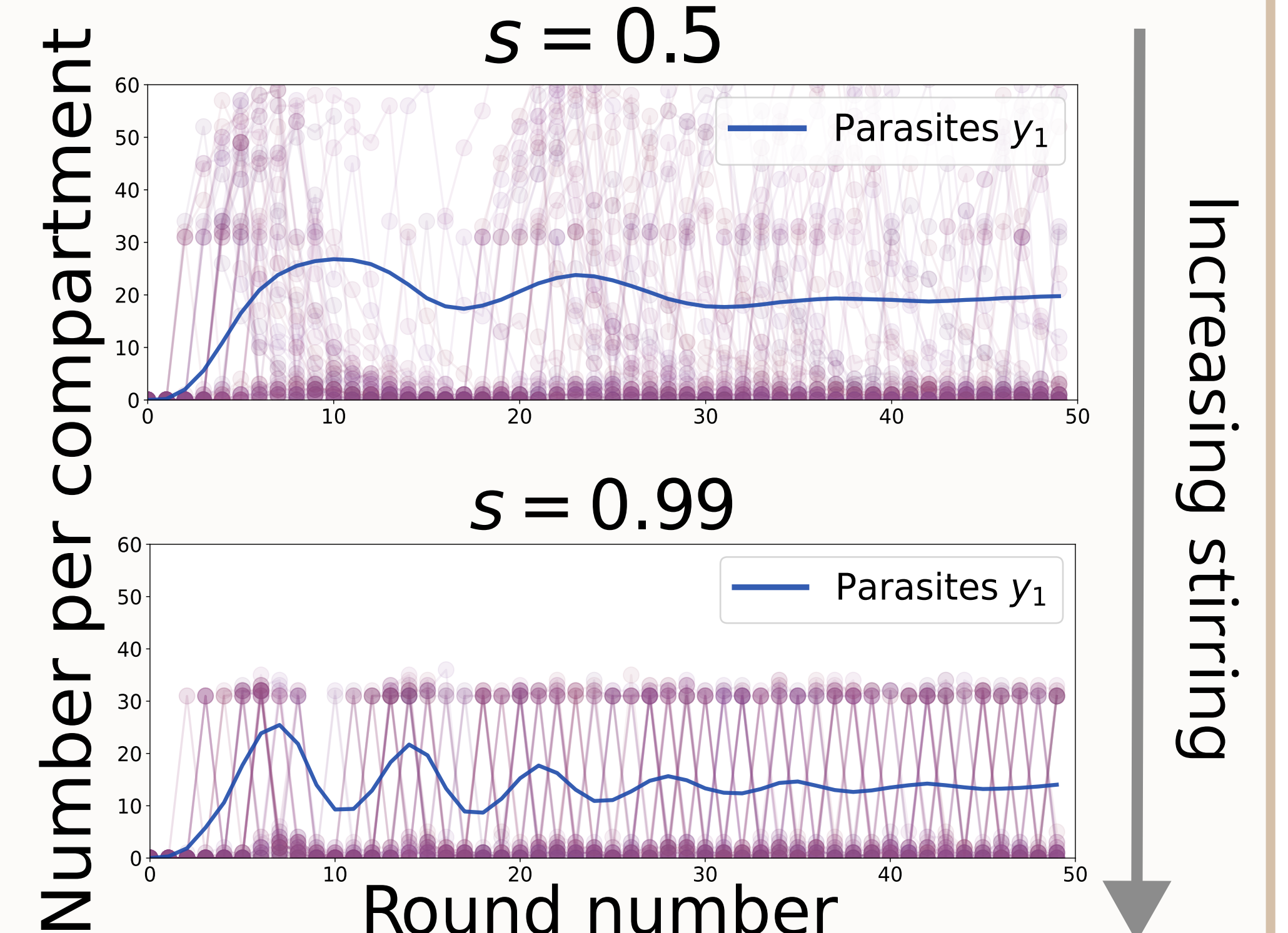
Green (G) and red (R) **fluorophores** are used.



Stirring is evaluated by assessing the **distribution of fluorescence**.

Composition at compartment level

A **weaker stirring** corresponds to an **heterogeneous population** and keeps a **compositional memory** of compartments [5]. **Stronger stirring** means **homogeneous** compartments (either filled with parasites or with replicases).

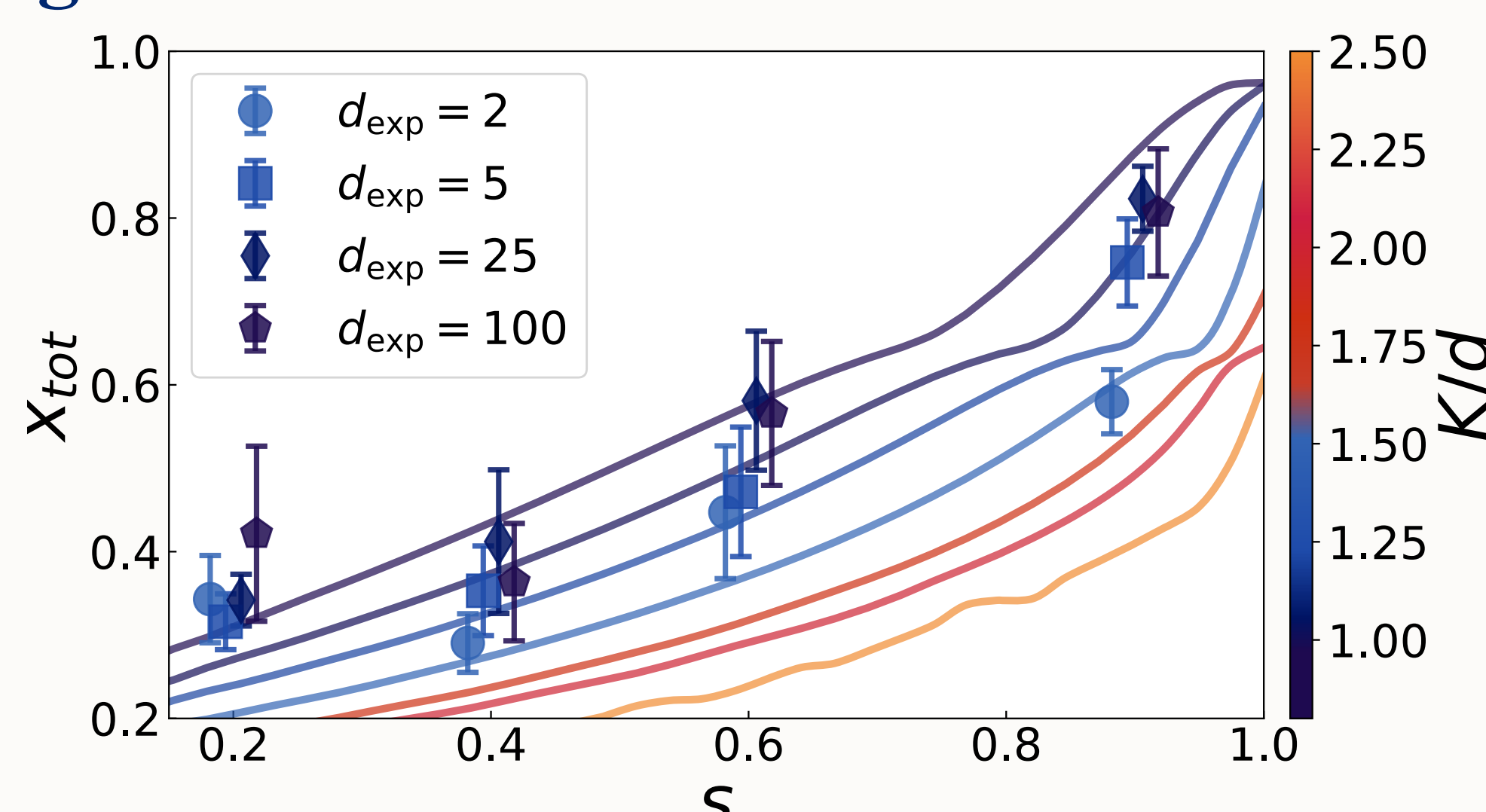


Stirring favors replicases

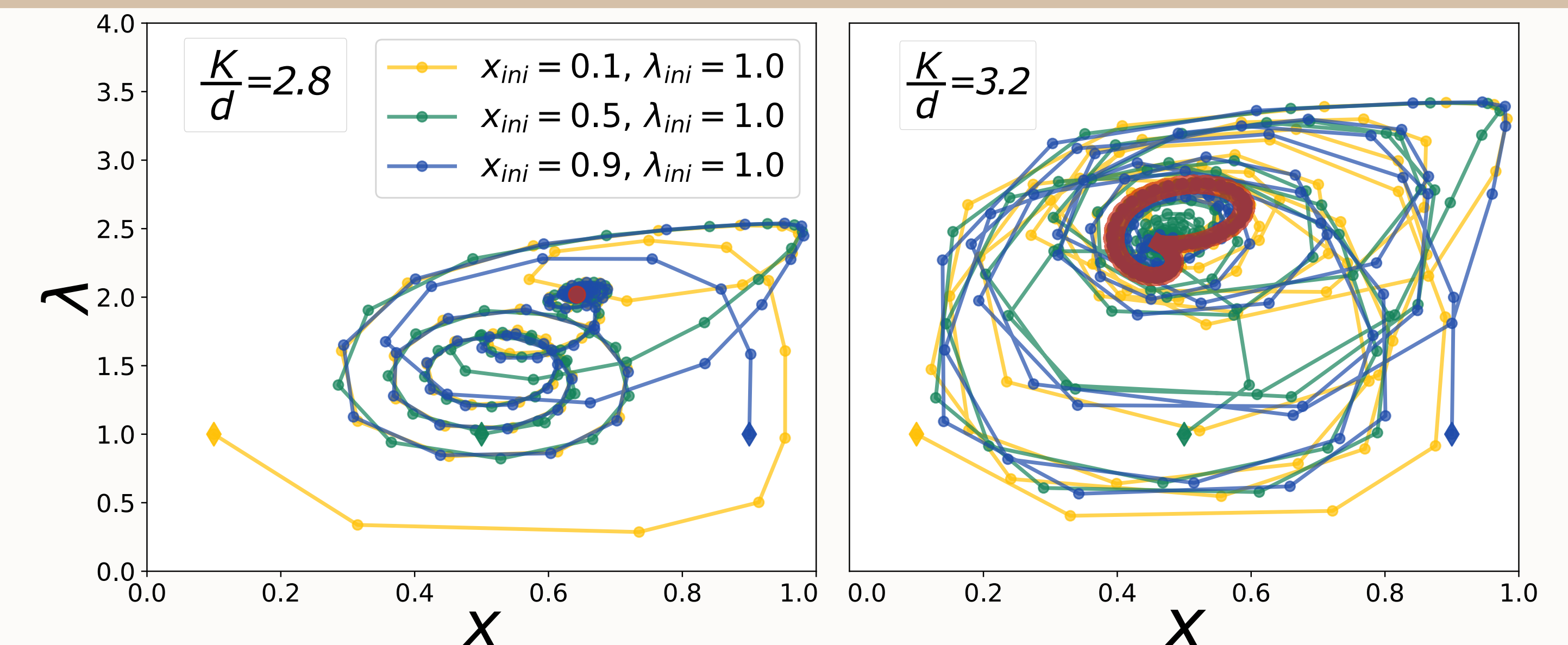
Stronger stirring and **small λ** isolate parasites, which **cannot replicate alone**

This leads to **increasing the fraction of replicases x_{tot}** as a function of **stirring s** .

Those trends are **validated experimentally**.



Limit cycles and stability



Oscillations in the plane (x, λ) where x is the **fraction of replicases** and λ the **number of individuals per compartment**. The ratio K/d determines if a **steady state** is reached.

[1]B. Ledoux,R. Kuwabara,N. Ichihashi,R. Mizuuchi, & D. Lacoste, Compositional memory matters for early molecular systems, Proc. Natl. Acad. Sci. (2026)

[2]R Mizuuchi, T Furubayashi, N Ichihashi, Evolutionary transition from a single RNA replicator to a multiple replicator network. Nat. Commun. (2022)

[3]A Blokhuis, D Lacoste, P Nghe, L Peliti, Selection Dynamics in Transient Compartmentalization. Phys. Rev. Lett. (2018)

[4]A Blokhuis, P Nghe, L Peliti, D Lacoste, The generality of transient compartmentalization and its associated error thresholds. J. Theor. Biol. (2020)

[5]M Wu, PG Higgs, Compositional Inheritance: Comparison of Self-assembly and Catalysis. Orig. Life Evol. Biospheres (2008).